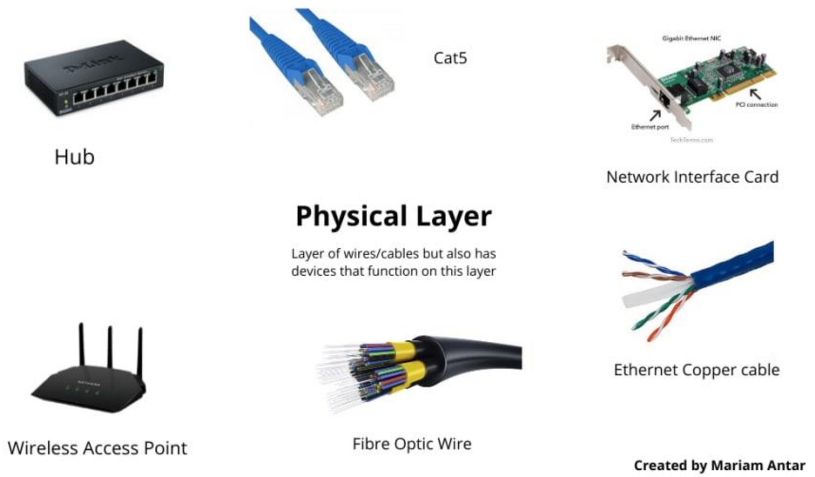
CHAPTER 4

Transmission Media

Physical Layer Characteristics

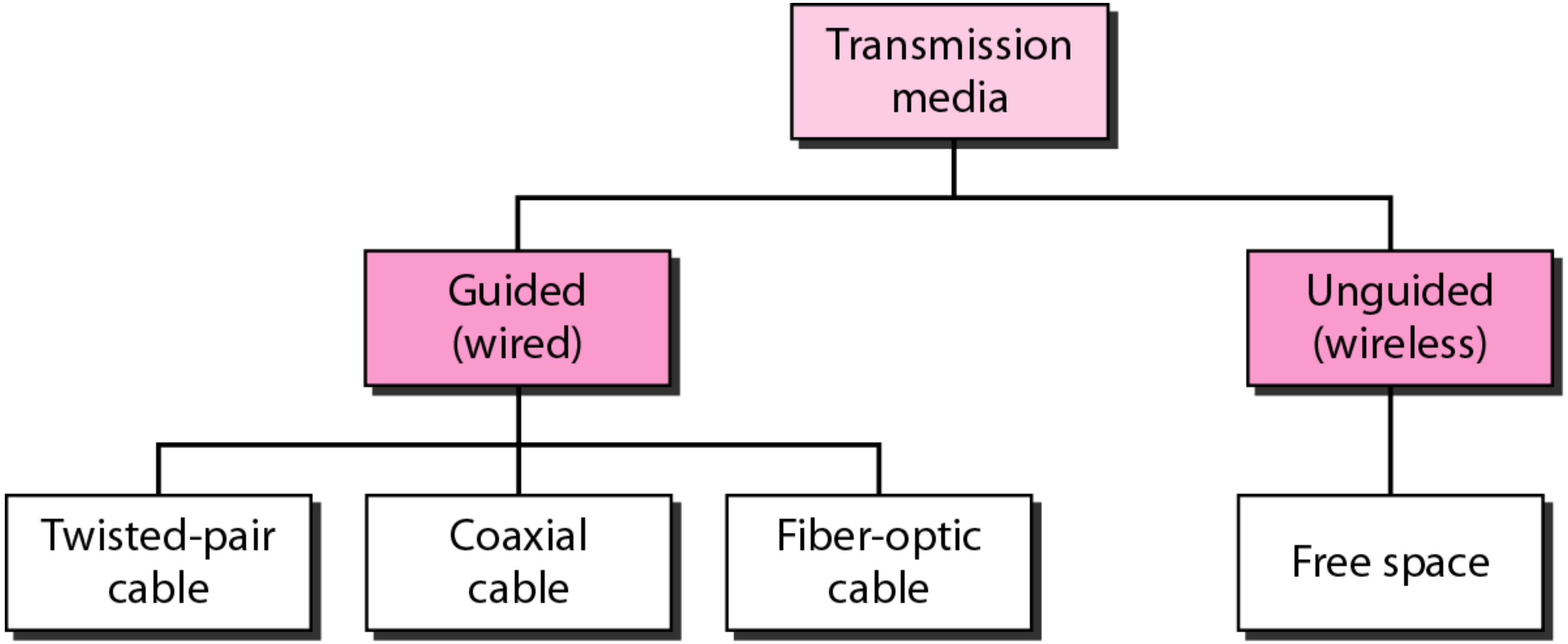
Physical Layer Standards address three functional areas:

- Physical Components
- Encoding
- Signaling



The Physical Components are the hardware devices, media, and other connectors that transmit the signals that represent the bits.

Classes of Transmission Media



Types of Copper Cabling

Unshielded Twisted-Pair (UTP) Cable

Shielded Twisted-Pair (STP) Cable

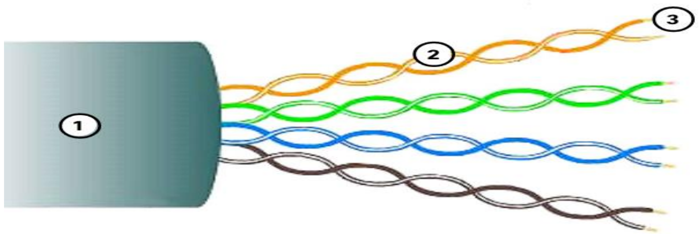
Coaxial Cable

Unshielded Twisted Pair (UTP)

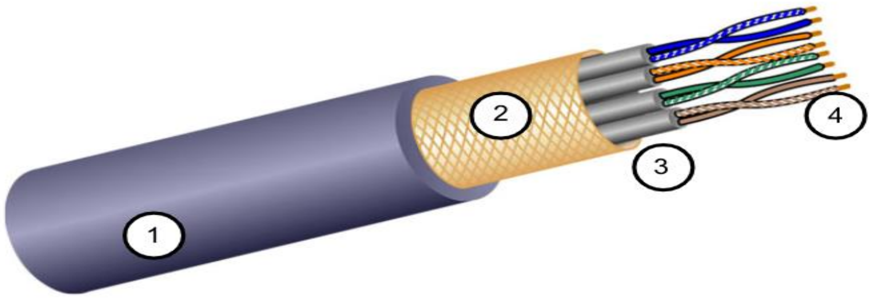
- UTP is the most common networking media.
- Terminated with RJ-45 connectors
- Interconnects hosts with intermediary network devices.

Key Characteristics of UTP

1. The outer jacket protects the copper wires from physical damage.
2. Twisted pairs protect the signal from interference.
3. Color-coded plastic insulation electrically isolates the wires from each other and identifies each pair.



Shielded Twisted Pair (STP)



- Better noise protection than UTP
- More expensive than UTP
- Harder to install than UTP
- Terminated with RJ-45 connectors
- Interconnects hosts with intermediary network devices

Key Characteristics of STP

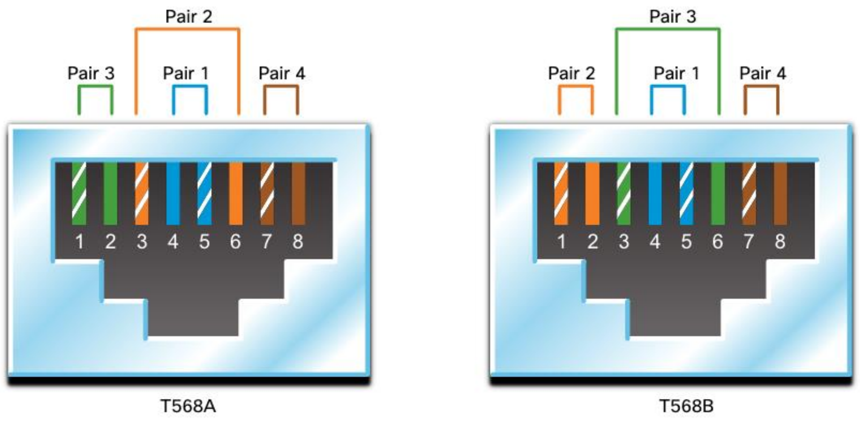
1. The outer jacket protects the copper wires from physical damage
2. Braided or foil shield provides EMI/RFI protection
3. Foil shield for each pair of wires provides EMI/RFI protection
4. Color-coded plastic insulation electrically isolates the wires from each other and identifies each pair

Properties of UTP Cabling

UTP has four pairs of color-coded copper wires twisted together and encased in a flexible plastic sheath. No shielding is used. UTP relies on the following properties to limit crosstalk:

- **Cancellation** - Each wire in a pair of wires uses opposite polarity. One wire is negative, the other wire is positive. They are twisted together, and the magnetic fields effectively cancel each other and outside EMI/RFI.
- **Variation** in twists per foot in each wire - Each wire is twisted a different amount, which helps prevent crosstalk amongst the wires in the cable.

Straight-through and Crossover UTP Cables

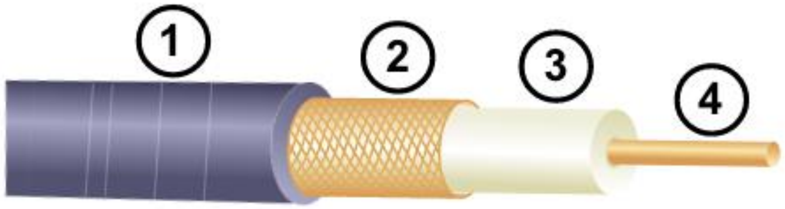


Cable Type	Standard	Application
Ethernet Straight-through	Both ends T568A or T568B	Host to Network Device
Ethernet Crossover *	One end T568A, other T568B	Host-to-Host, Switch-to-Switch, Router-to-Router
* Considered Legacy due to most NICs using Auto-MDIX to sense cable type and complete connection		

Coaxial Cable

Consists of the following:

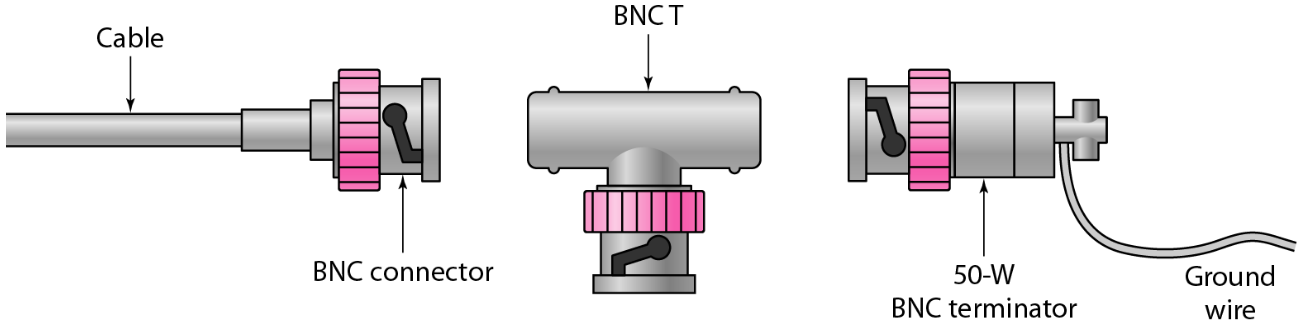
1. Outer cable jacket to prevent minor physical damage
2. A woven copper braid, or metallic foil, acts as the second wire in the circuit and as a shield for the inner conductor.
3. A layer of flexible plastic insulation
4. A copper conductor is used to transmit the electronic signals.



Categories of Coaxial Cables

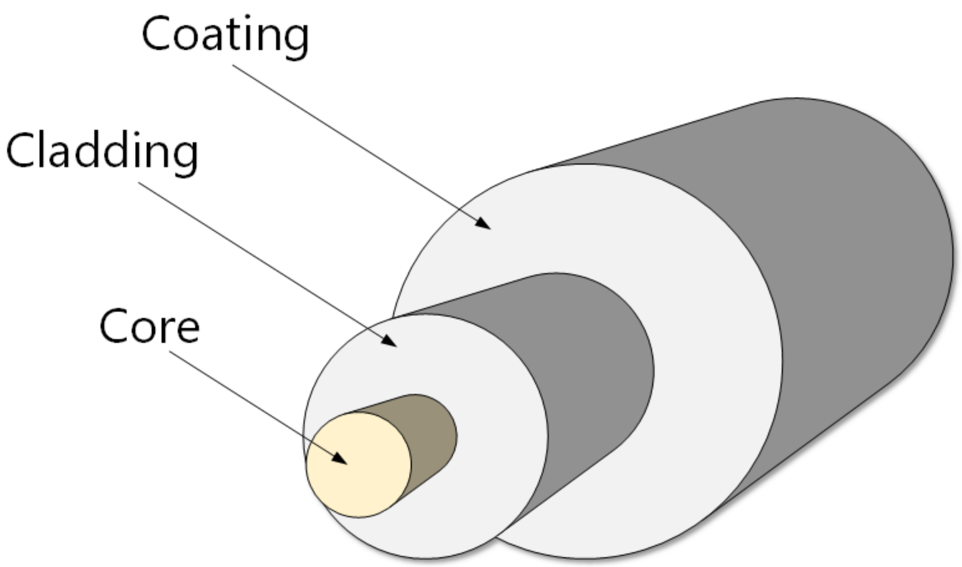
Category	*Impedance*	*Use*
RG-59	75 Ω	Cable TV
RG-58	50 Ω	Thin Ethernet
RG-11	50 Ω	Thick Ethernet

BNC connectors



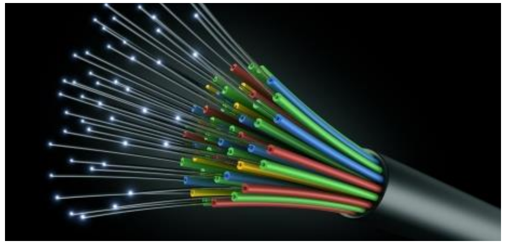
Optical Fiber

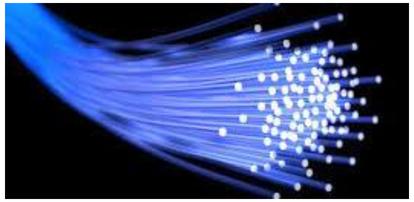
A light source shines a light into the **core**. The core is surrounded by optical material called the **cladding** that keeps the light in the core using an optical technique called *total internal reflection*. Outer ***coating*** layer protects the interior and makes the cable stronger and easier to install and manage.



Properties of Fiber-Optic Cabling

- Transmits data over longer distances at higher bandwidth than any other networking media
- Less susceptible to attenuation, and completely immune to EMI/RFI
- Made of flexible, extremely thin strands of very pure glass
- Uses a laser or LED to encode bits as pulses of light
- The fiber-optic cable acts as a wave guide to transmit light between the two ends with minimal signal loss



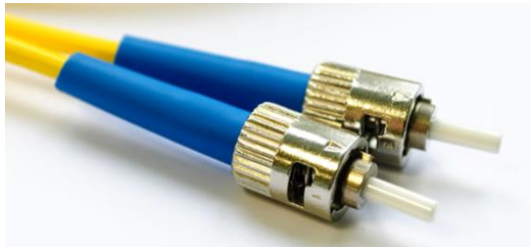


Fiber-Optic Cabling Usage

Fiber-optic cabling is now being used in four types of industry:

1. **Enterprise Networks -** Used for backbone cabling applications and interconnecting infrastructure devices
2. **Fiber-to-the-Home (FTTH) -** Used to provide always-on broadband services to homes and small businesses
3. **Long-Haul Networks -** Used by service providers to connect countries and cities
4. **Submarine Cable Networks -** Used to provide reliable high-speed, high-capacity solutions capable of surviving in harsh undersea environments at up to transoceanic distances.

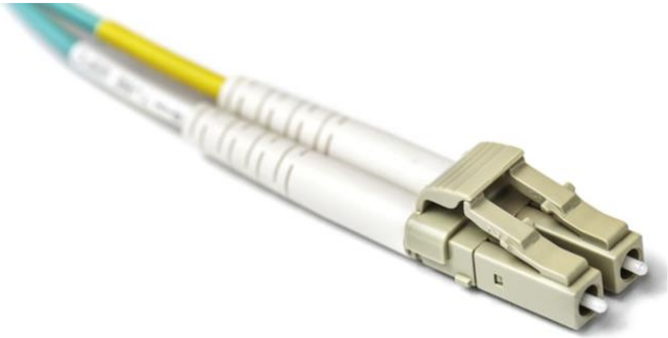
Fiber-Optic Connectors



Straight-Tip (ST) Connectors

Lucent Connector (LC) Simplex Connectors

Subscriber Connector (SC) Connectors



Duplex Multimode LC Connectors

Fiber Patch Cords

SC-SC MM Patch Cord

LC-LC SM Patch Cord

ST-LC MM Patch Cord

ST-SC SM Patch Cord

A yellow jacket is for single-mode fiber cables and orange for multimode fiber cables.

Fiber versus Copper

Implementation Issues	UTP Cabling	Fiber-Optic Cabling
Bandwidth supported	10 Mb/s - 10 Gb/s	Up to 100 Gb/s or more
Distance	Relatively short (1 - 100 meters)	Relatively long (1 - 100,000 meters)
Immunity to EMI and RFI	Low	High
Media and connector costs	Lowest	Highest
Installation skills required	Lowest	Highest

Advantages and Disadvantages of Optical Fiber

Advantages

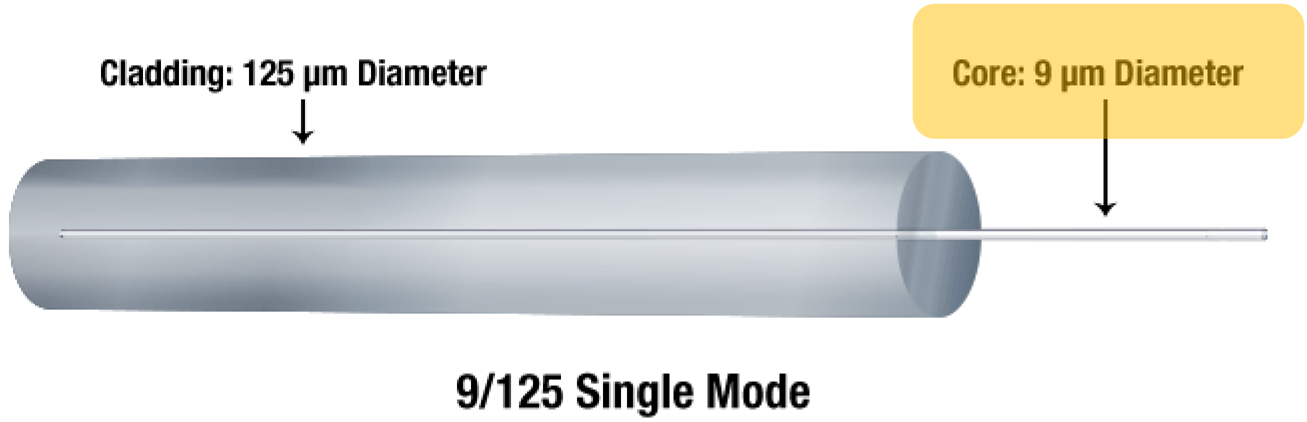
- High bandwidth
- Long distances
- Immunity to interference
- Smaller size and lighter weight
- Security
- Durability

Disadvantages

- Cost
- Installation
- Complexity
- Fragility

Propagation Modes: Single Mode

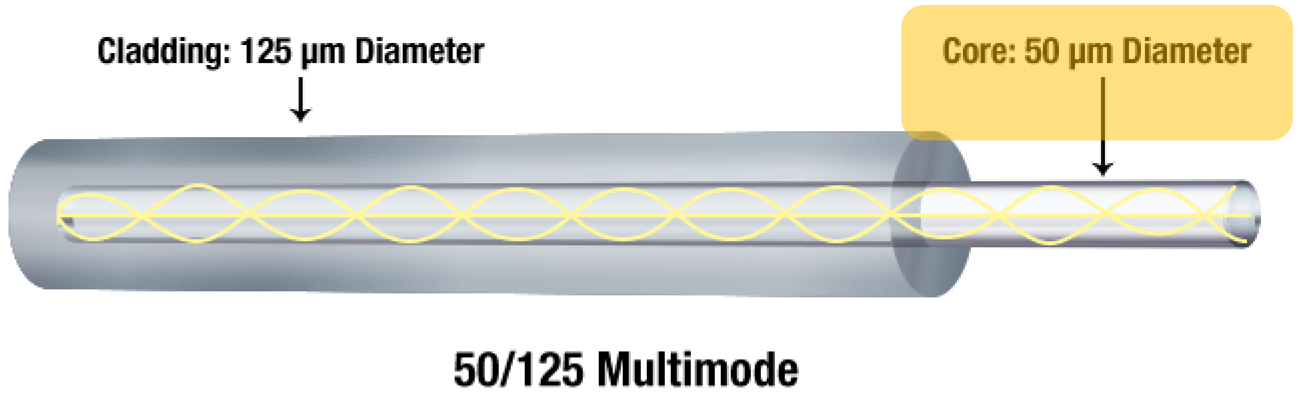
The light is traveling straightway as the core of the diameter is very less.



Single-Mode Fiber (SMF)

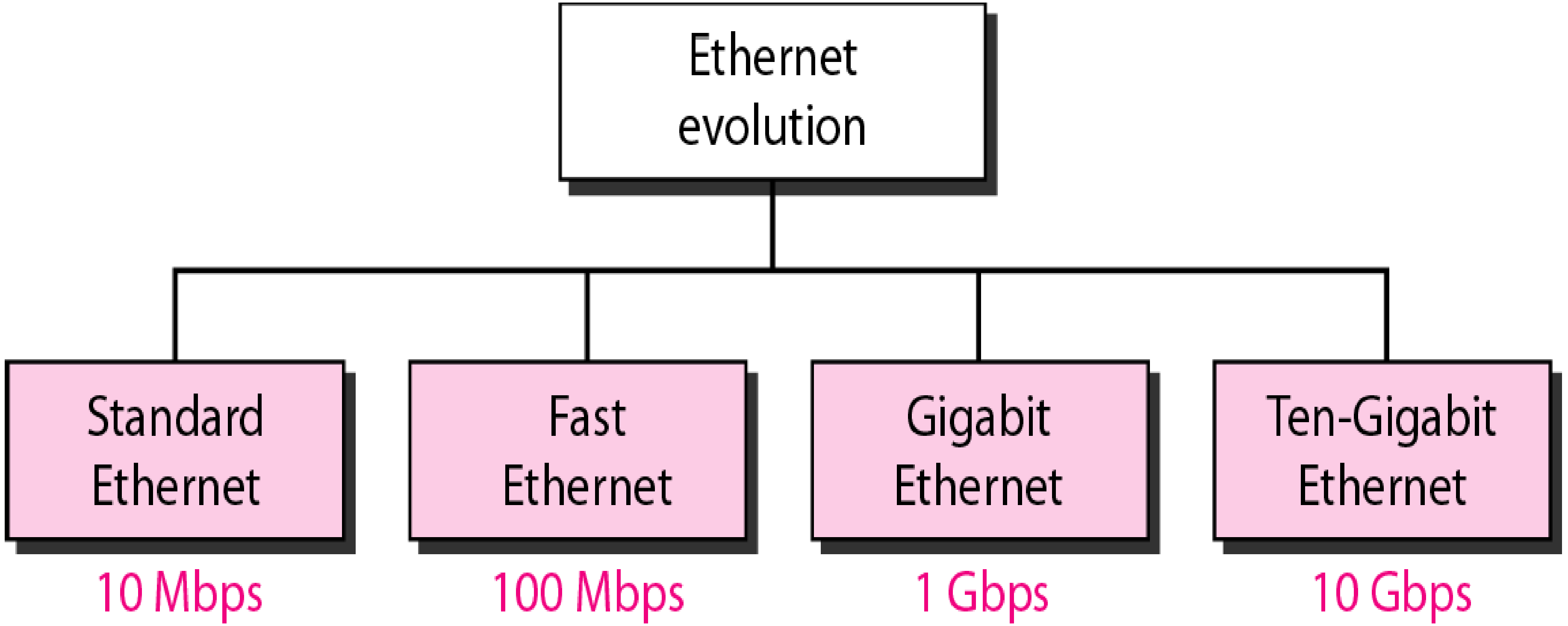
Propagation Modes: Multimode

Multimode fiber optic cable has a large diameter core, so multiple modes of light propagate in it. It has more data carrying capacity due to multiple reflection inside.



Multimode fiber (MMF).

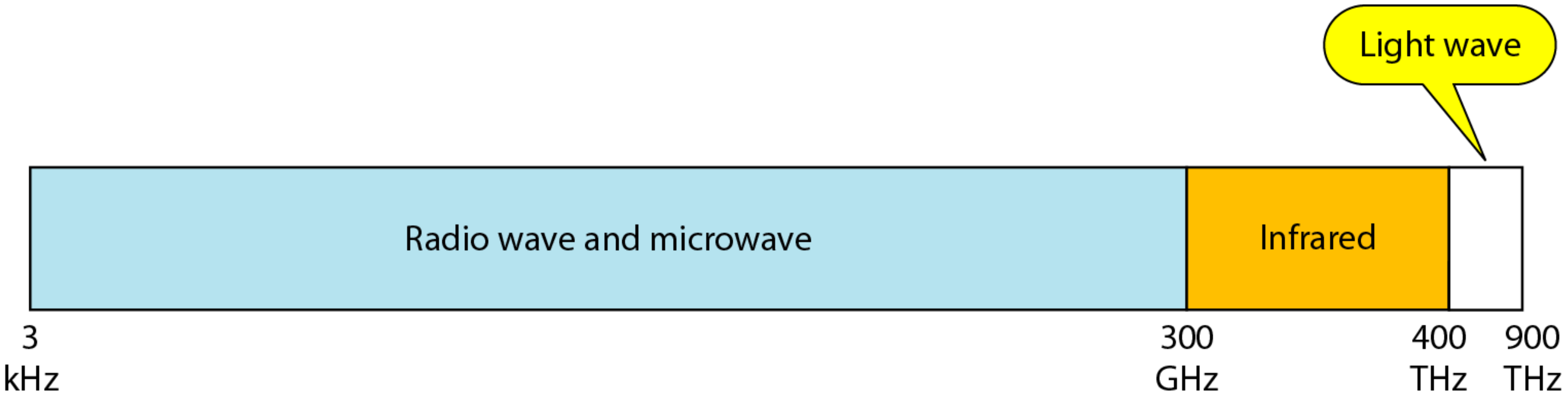
Ethernet evolution through four generations



Wi-Fi, WiMAX, Bluetooth, ZigBee, LiFi

Unguided Media: Wireless

Unguided media transport electromagnetic waves without using a physical conductor. This type of communication is often referred to as wireless communication.



Properties of Wireless Media

It carries electromagnetic signals using radio or microwave frequencies.

Some of the limitations of wireless:

- **Coverage area** - Effective coverage can be impacted by the physical characteristics of the deployment location.
- **Interference** - Wireless is susceptible to interference and can be disrupted by many common devices.
- **Security** - Wireless communication coverage requires no access to a physical strand of media, so anyone can gain access to the transmission.

Wireless Standards

The IEEE and telecommunications industry standards for wireless data communications cover both the data link and physical layers.

Wireless Standards:

- **Wi-Fi (IEEE 802.11)** - Wireless LAN (WLAN) technology
- **Bluetooth (IEEE 802.15)** - Wireless Personal Area network (WPAN) standard
- **WiMAX (IEEE 802.16)** - Uses a point-to-multipoint topology to provide broadband wireless access
- **Zigbee (IEEE 802.15.4)** - Low data-rate, low power-consumption communications, primarily for Internet of Things (IoT) applications

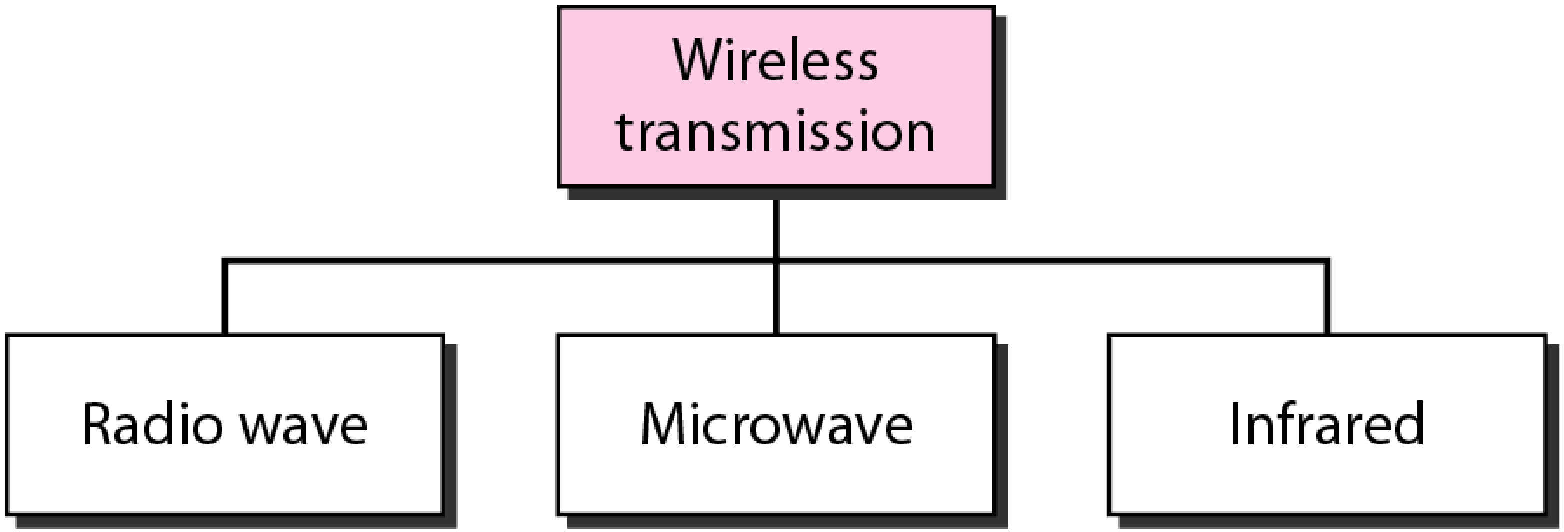
Wireless LAN

In general, a Wireless LAN **(WLAN) requires the following** devices:

- **Wireless Access Point** **(AP)** - Concentrate **wireless signals** from users and connect to the **existing copper-based** network infrastructure
- **Wireless NIC Adapters** - Provide wireless communications capability to network hosts

Network Administrators must develop and apply stringent security policies and processes to protect WLANs from unauthorized access and damage.

Wireless Transmission Waves



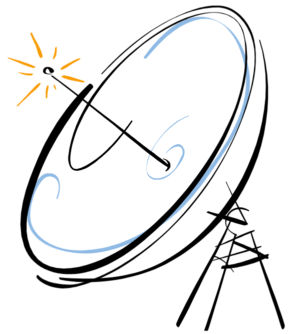
Wireless Transmission Waves

- Radio waves are used for **multicast** communications, such as radio and television. They can penetrate through walls. Use **omni** directional antennas.

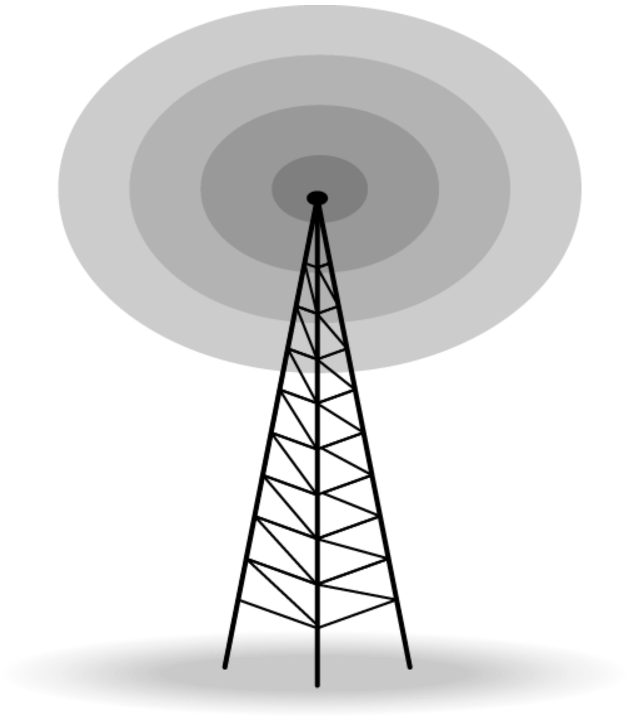
- Microwaves are used for **unicast** communication such as cellular telephones, satellite networks, and wireless LANs. Higher frequency ranges cannot penetrate walls. Use **directional** antennas - *point to point line of sight communications.*

- Infrared signals can be used for **short-range** communication in a closed area using **line-of-sight** propagation.

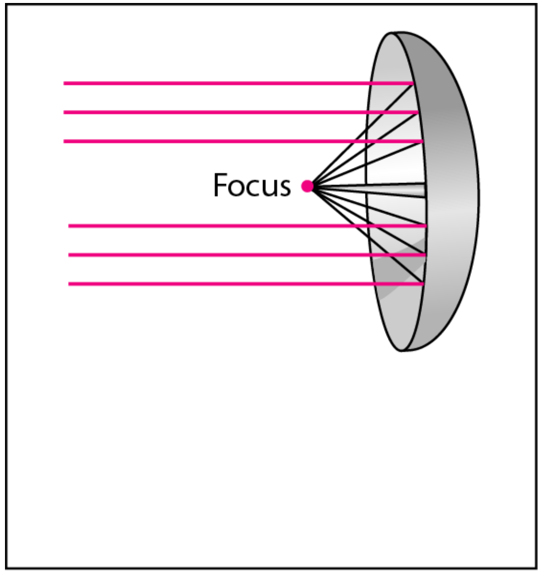
Antennas



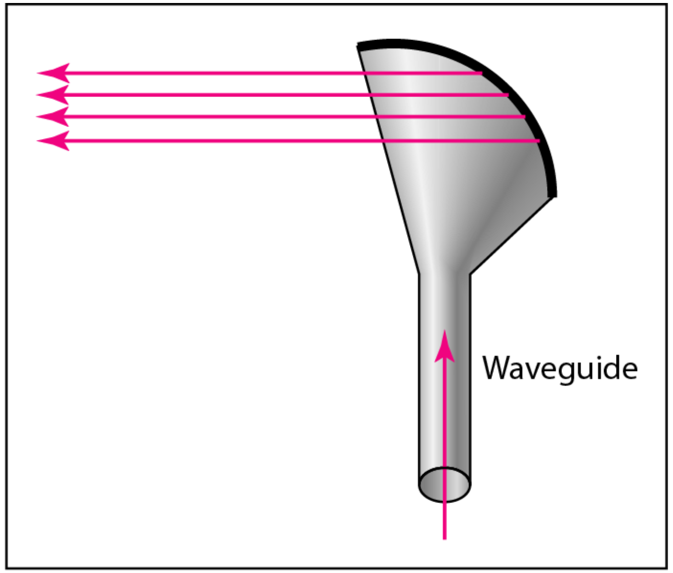
- Electrical conductor used to radiate or collect electromagnetic energy
- **Transmitter** Radio frequency electrical energy is converted into electromagnetic energy by the antenna and radiated into the surrounding environment
- **Reception** occurs when the electromagnetic signal intersects the antenna
- In two-way communication, the same antenna can be used for both transmission and reception



Omnidirectional antenna



a. Dish antenna

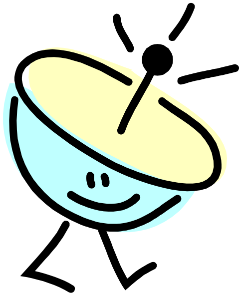


b. Horn antenna

Unidirectional antennas

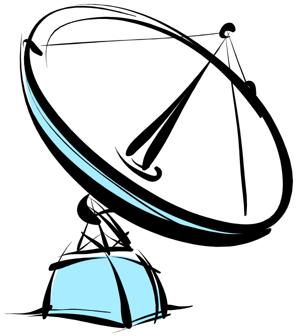
Antenna Gain

- A measure of the directionality of an antenna
- Defined as the power output versus that produced by an antenna
- Measured in decibels (dB)
- Effective area of an antenna is related to the physical size of the antenna and to its shape



Terrestrial Microwave

- Most common type is the parabolic "dish", size is about 3 m in diameter
- Antenna is fixed rigidly and focuses a narrow beam to achieve line-of-sight transmission to the receiving antenna
- Usually located at substantial heights above ground level
- A series of microwave relay towers is used to achieve long-distance transmission

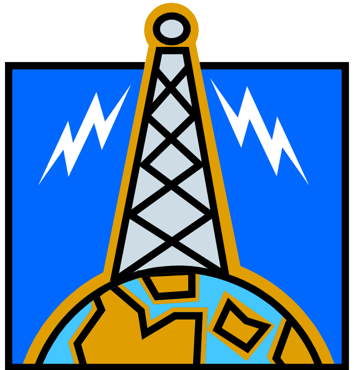


Terrestrial Microwave Applications

- Used for long telecommunications service as an alternative to coaxial cable or optical fiber
- Used for both voice and TV transmission
- 1-40GHz frequencies, with higher frequencies having higher data rates
- Main source of loss is attenuation caused mostly by distance, rainfall and interference

Satellite Microwave

- A communication satellite is in effect a microwave relay station
- Used to link two or more ground stations
- Receives transmissions on one frequency band, amplifies or repeats the signal, and transmits it on another frequency



Satellite Microwave Applications

- Most important applications for satellites are:

Long-distance telephone transmission

Private business networks

Television distribution

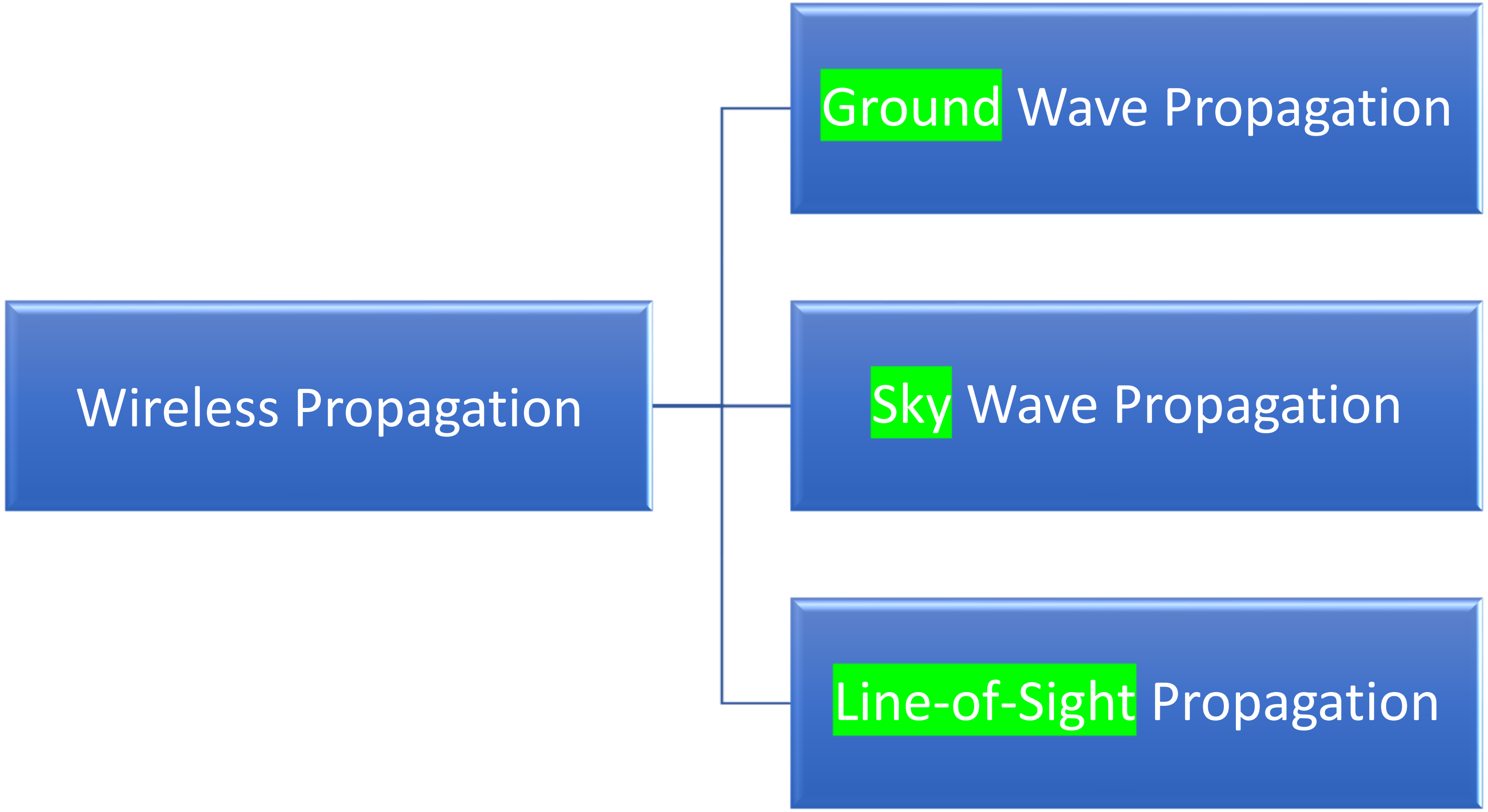
Global Positioning System (GPS)

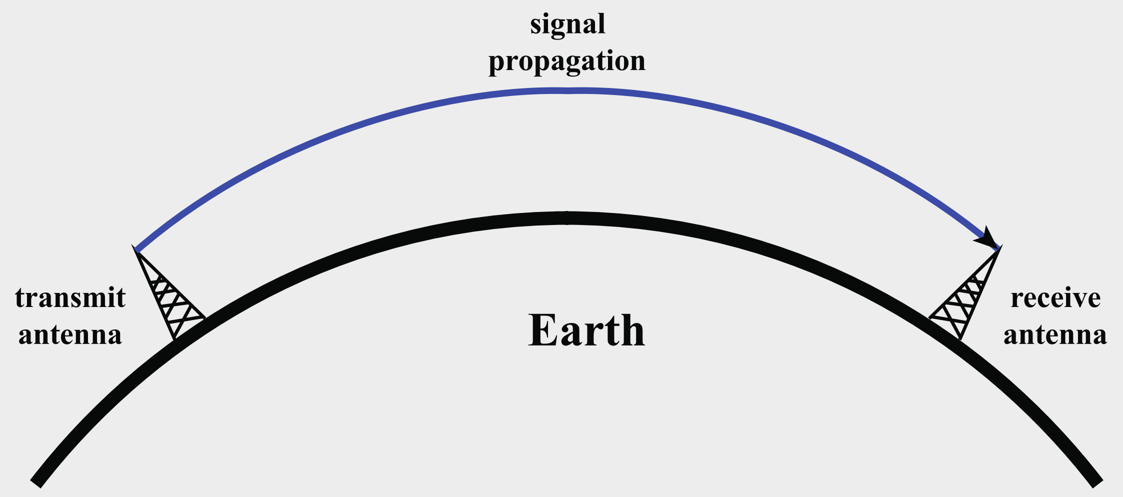
Transmission Characteristics

- The optimum frequency range for satellite transmission is 1 to 10 GHz

- Satellites use a frequency bandwidth range of 5.925 to 6.425 GHz from earth to satellite (uplink) and a range of 3.7 to 4.2 GHz from satellite to earth (downlink)

Wireless Propagation

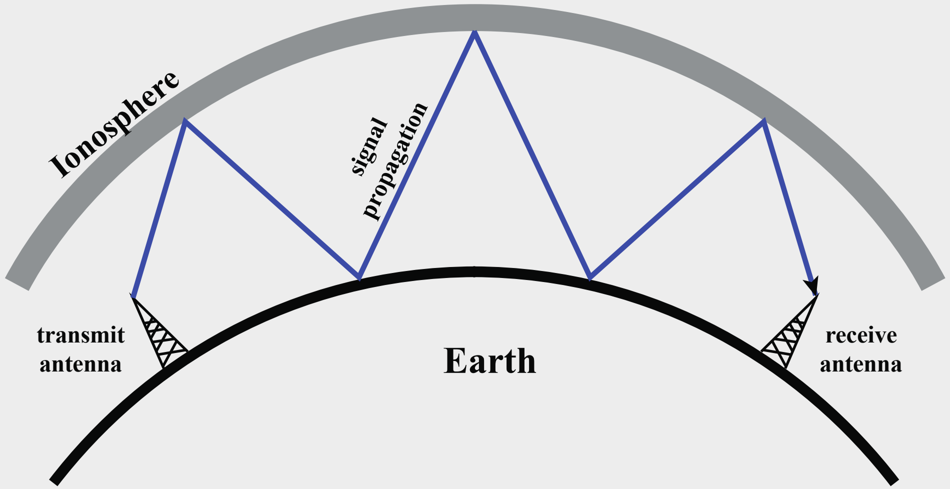




(a) Ground-wave propagation (below 2 MHz)

Figure 4.11 Wireless Propagation Modes

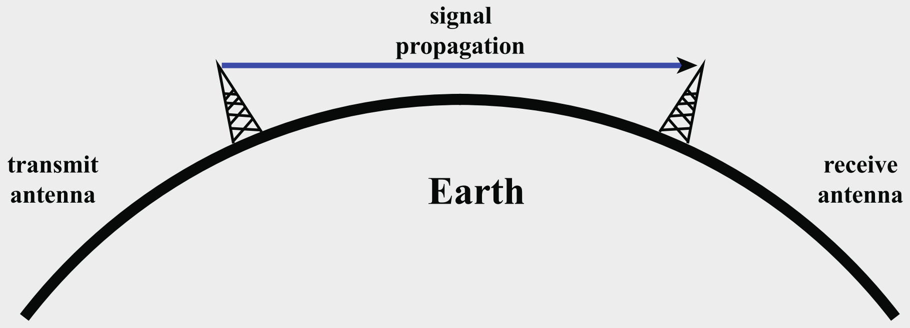
- Ground wave propagation follows the contour of the earth
- The best-known example of ground wave communication is AM radio



(b) Sky-wave propagation (2 to 30 MHz)

Figure 4.11 Wireless Propagation Modes

- A signal from an earth-based antenna is reflected from the ionized layer of the upper atmosphere back down to earth
- Sky wave signals can travel through a number of hops, bouncing back and forth between the ionosphere and the earth’s surface



(c) Line-of-sight (LOS) propagation (above 30 MHz)

Figure 4.11 Wireless Propagation Modes

The term optical line of sight refers to the straight-line propagation with no obstacles of light waves

The propagation of radio waves bent by the curvature of the earth.